

Computer vision

Colour

Colour

<http://vgg.fiit.stuba.sk/kniha/>

Colour measurement

important for many applications

textile industry

paper industry

leather industry

graphics arts industry

...

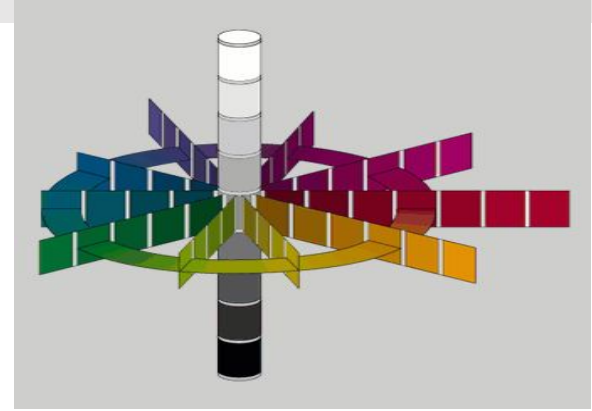
standardized

Color Atlas historical milestones

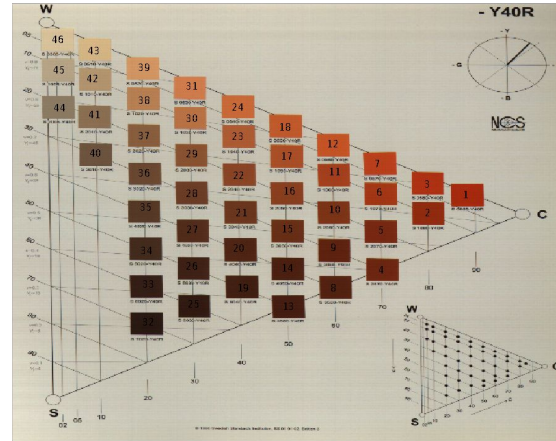
Munsell Color Atlas (1913)

Munsell notation
for a chromatic color is written
symbolically: H V/C. For example,
5.3R 6.1/14.4.

Munsell color space



NCS Natural Color System – SIS
Standardiseringen i Sverige



Radiometry vs. Photometry

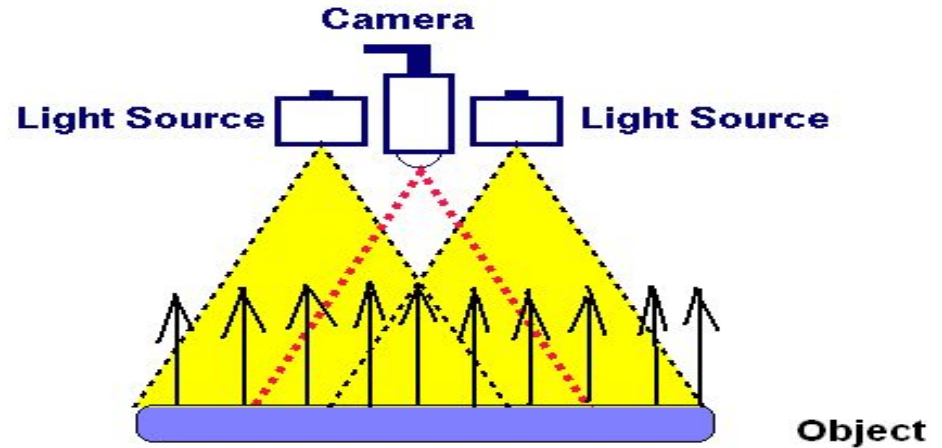
Radiometry is the science and technology of the measurement of **electromagnetic radiation** over the whole spectrum of electromagnetic waves.

Photometry is the science of the measurement of light, in terms of its **perceived brightness to the human eye**.

Image Sensing and Acquisition

Image Sensing and Acquisition

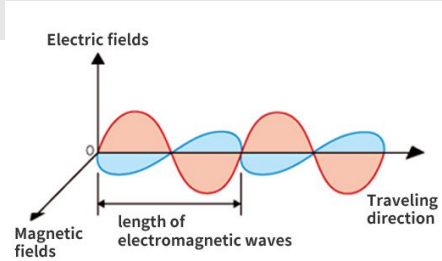
Depending on the nature of the source, illumination energy is reflected from, or transmitted through, objects.



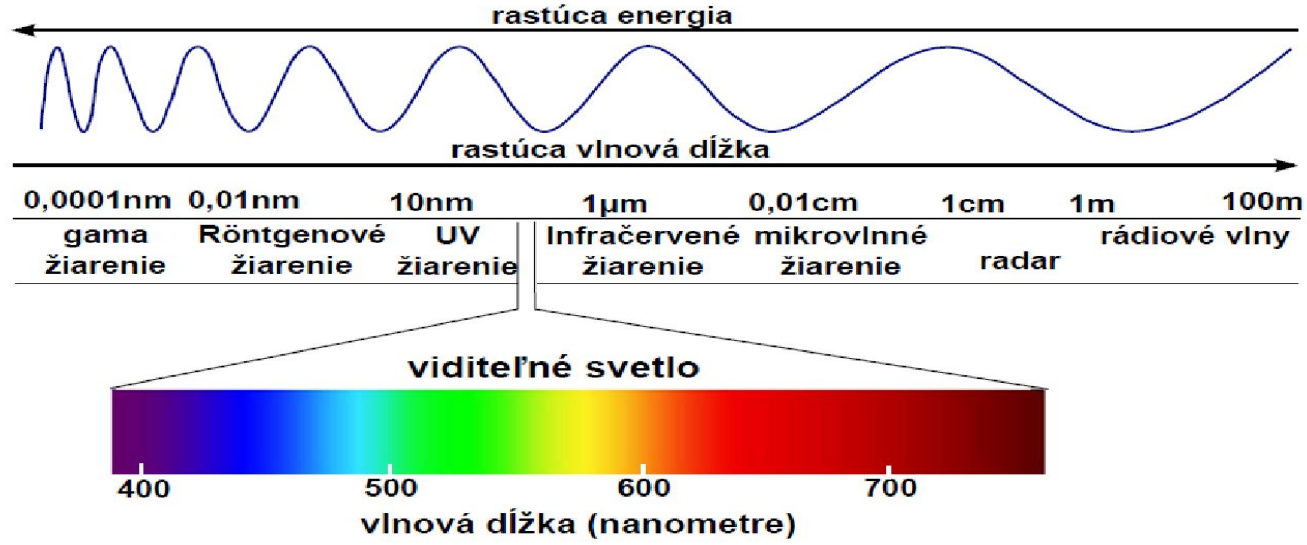
1. Light source
2. Object reflection
3. Sensor

Spectrum of light source

Electromagnetic spectrum



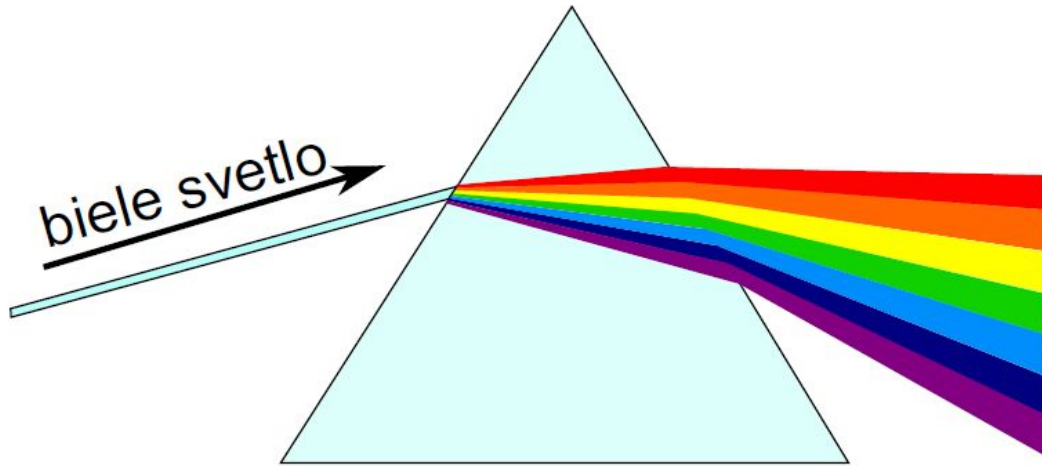
Electromagnetic wave



Visible range of the electromagnetic spectrum

Splitting White Light to Monochromatic Colors

Light Prism



The spectrum of colors produced by a prism is referred to as spectrally pure or monochromatic - they are related to the wavelength.

Different spectrally pure colors are said to have a different hue.

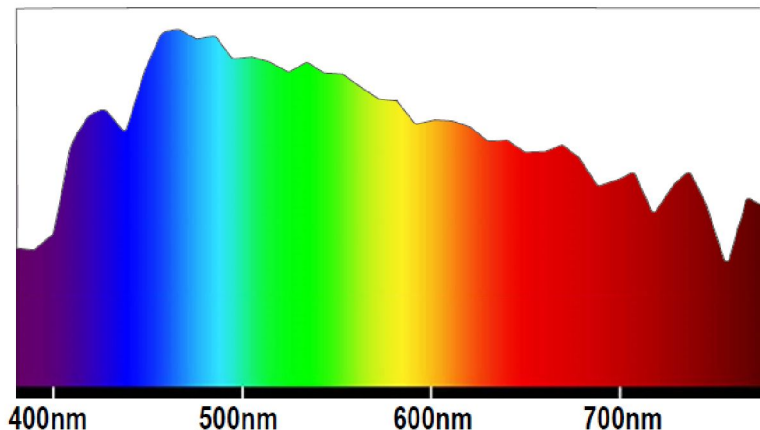
A spectrally pure or monochromatic color can be produced by a single wavelength.

Spectral power distribution (SPD)

Representation of the radiant power emitted by a light source at each wavelength or band of wavelengths in the visible region of the electromagnetic spectrum (360 to 770 nm).

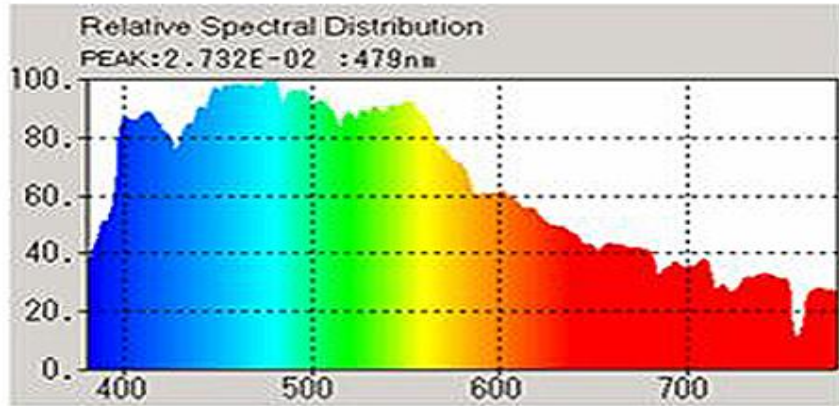
SPD describes how the power of a signal is distributed with frequency

SPD: $S(\lambda)$

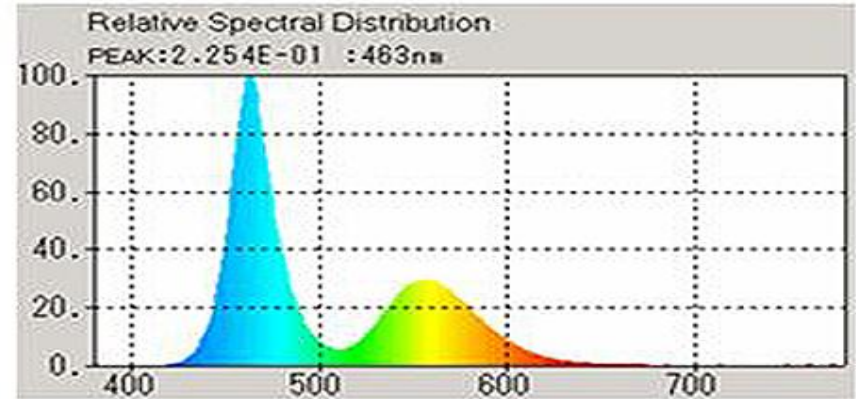


SPD curves of specific light sources

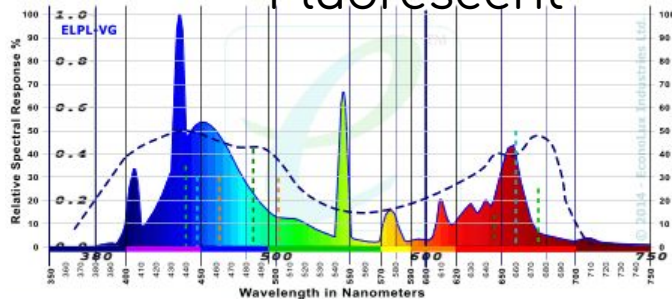
Daylight



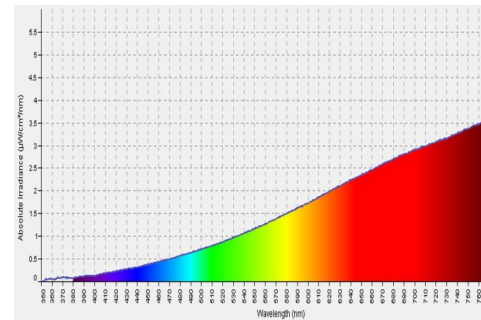
White LED



Fluorescent



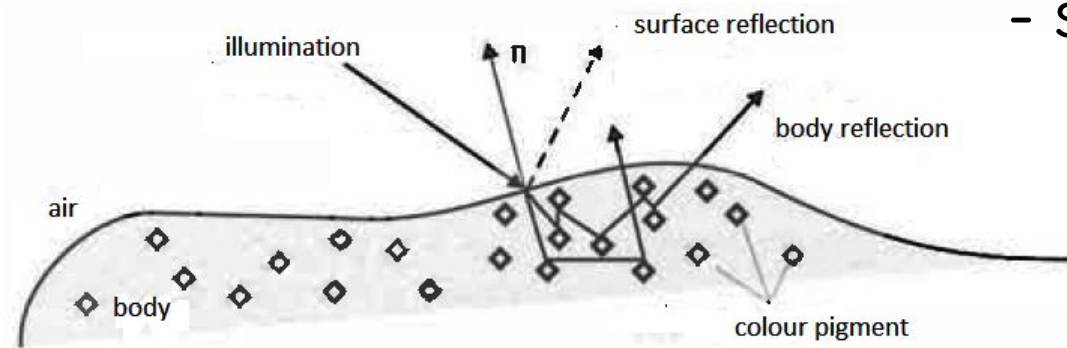
Incandescent light bulb



Reflectance/transmittance
of the object

Surface reflection

Observed colour of objects is caused by certain wavelength absorptions by pigment particles in dielectrics.

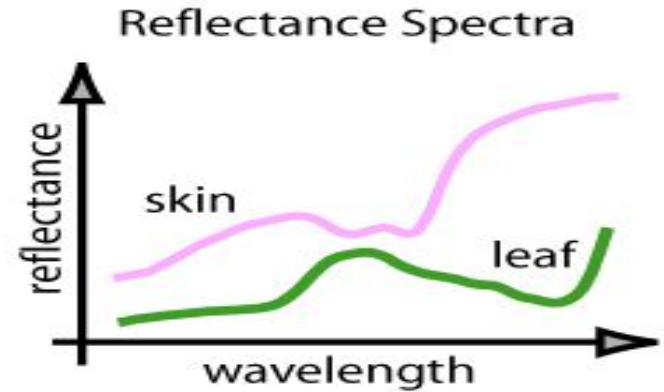


- specular component

Reflectance spectrum of a surface

Colour of a surface depends on the **reflectance spectrum** $r(\lambda)$ and on the **illuminance spectrum** $i(\lambda)$

Range of the reflectance spectrum: 0-1



reflectance spectra of human skin and a green leaf.

Reflection of light from a surface $R(\lambda)$

The function $R(\lambda)$ may be characterized by a multiplication of two components **illumination** $i(\lambda)$ and **reflectance** $r(\lambda)$:

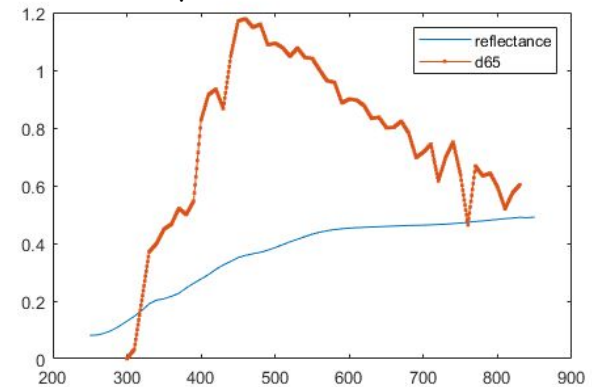
$$R(\lambda) = \text{illumination } i(\lambda) * \text{reflectance } r(\lambda)$$

$$R(\lambda) = i(\lambda) * r(\lambda)$$

$r(\lambda)$ - reflectance spectrum

$i(\lambda)$ - illuminance spectrum

Example:



Reaction of a photoreceptor

Let:

$I(\lambda)$ be the spectral density of the illumination - illumination spectrum

$S(\lambda)$ be the spectral sensitivity of the sensor

$R(\lambda)$ be the reflectance spectrum -> describe how the surface patch reflects each wavelength of the illuminating light - spectral sensitivity of the surface

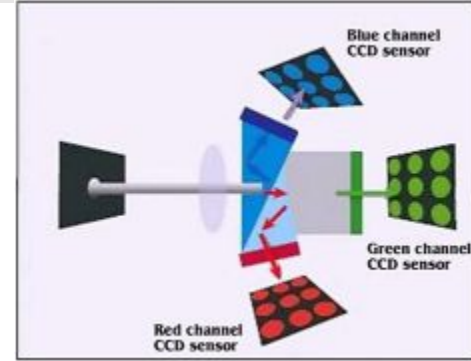
The spectral response q_i of the i -th sensor band (typically $i=3$), can be modelled by integration over a certain range of wavelengths $\lambda_1 - \lambda_2$:

$$Q = \int_{\lambda_1}^{\lambda_2} I(\lambda) * S(\lambda) * R(\lambda)$$

Optical Color Sensor

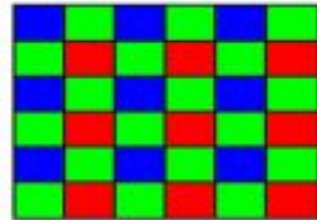
Colour camera

Three-chip cameras have 3 sensors prism each of the red, green and blue filters covers a complete sensor chip.



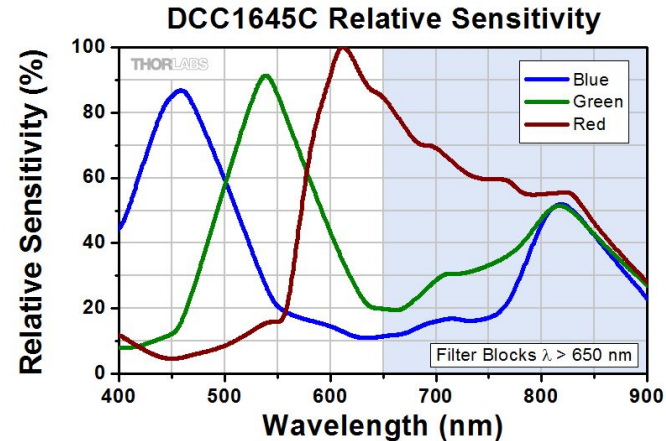
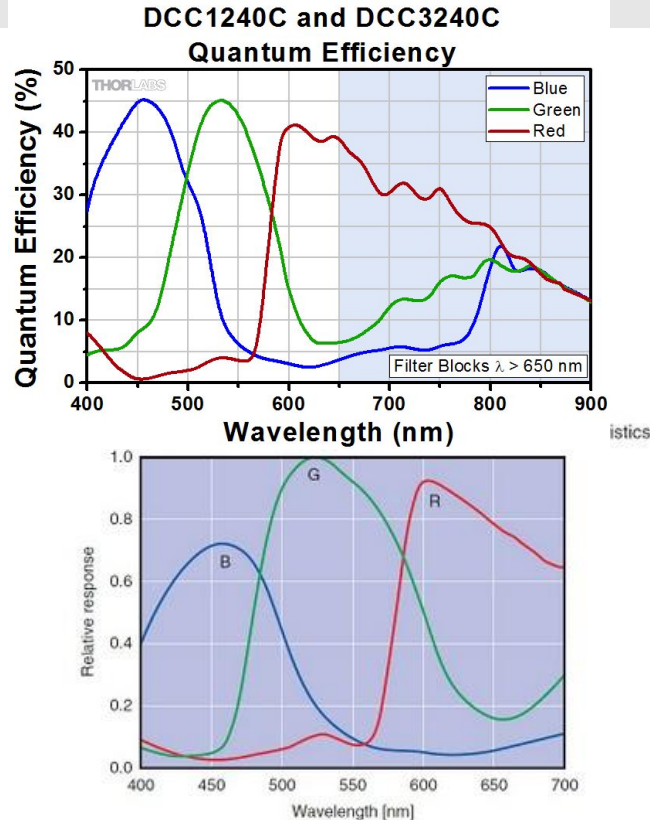
Single-chip colour sensors are conventional sensors to which a colour filter is applied.

Bayer matrix



For cost reasons, however, single-chip color cameras are still more widely used than three-chip cameras.

Spectral sensitivity of colour camera



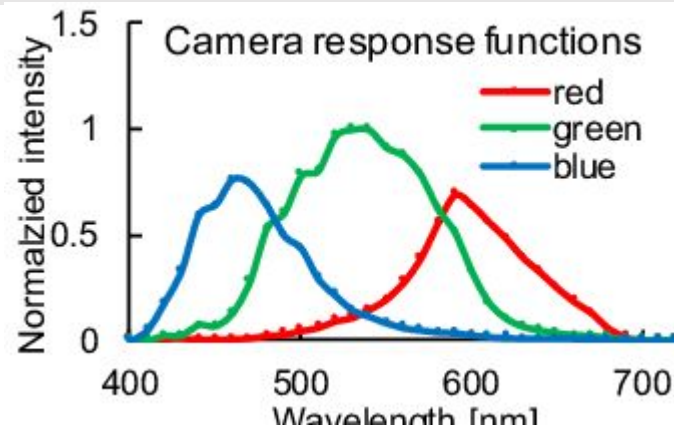
CMOS sensor
SONY camera

Multispectral images

Most sensors used for colour capture, e.g., in cameras (typically 3 channels), do not have accurate information about the colour

-> **spectrophotometer** -> Multispectral image

Each spectral band (typically 5nm or 10nm) is digitized independently and is represented by an individual digital image function as if it were a monochromatic image.



Example:

Spectral response function of Canon EOS 50D Digital Single-Lens Reflex (DSRL) camera

Multispectral images

Multispectral images with are commonly used in **remote sensing** from satellites, airborne sensors and in industry.

...for instance, the LANDSAT satellite transmits digitized images in 3-11 spectral bands from near-ultraviolet to infrared.

Color perceived by a human

Colour of reflected light / Colour perceived

Colour of reflected light depends on

- Spectral radiance of light falling on that patch
- Spectral reflectance of the surface light is leaving

Colour perceived by human depends additionally on

- Visual system receptors and brain processing, environment

Colour (perceived by human)

Color is the **perceptual result** of light in the visible region of the spectrum, having wavelengths in the region of 400 nm to 700 nm.

Color stimuli are assumed to be uniquely defined by their radiant **spectral power distribution (SPD)** -power spectral distributions of $P(\lambda)$
(often in x components each representing a 10 nm(5nm) band).

Human colour perception adds a subjective layer on top of underlying objective physical properties .

The Eye

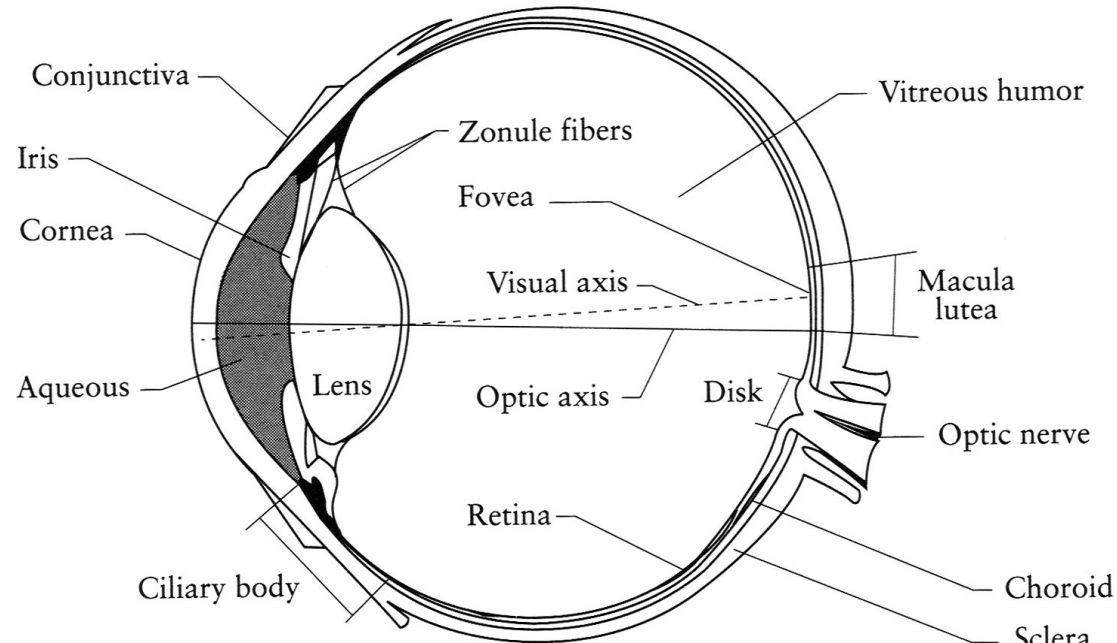
The human eye is like a camera!

Iris (*duhovka*) - colored annulus with radial muscles

Pupil (*zornice*) - the hole (aperture) whose size is controlled by the iris

Lens (*čočka*) - changes shape by using ciliary muscles (to focus on objects at different distances)

Retina (*sítnice*) - photoreceptor cells



Photoreceptors in human eye

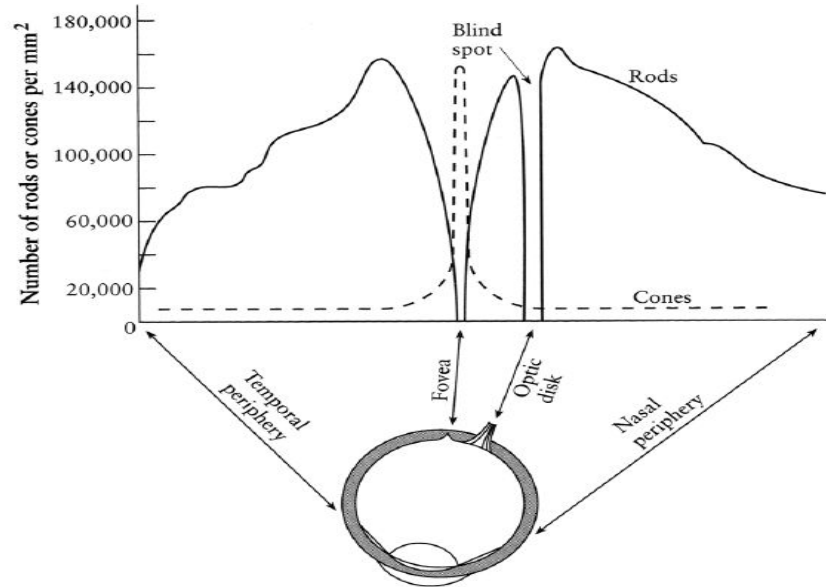
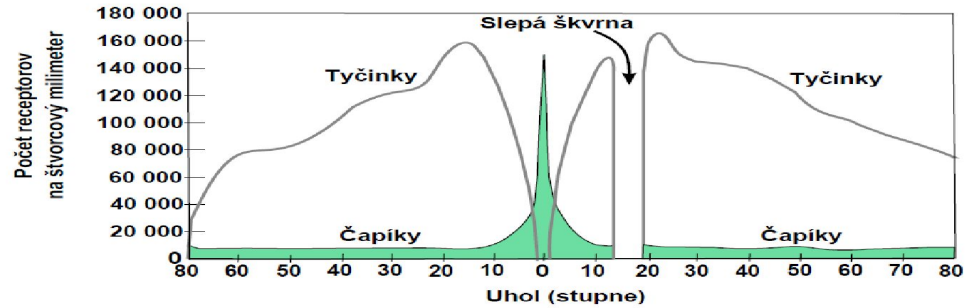
Rods & Cones

Cones (čípky sítnice)

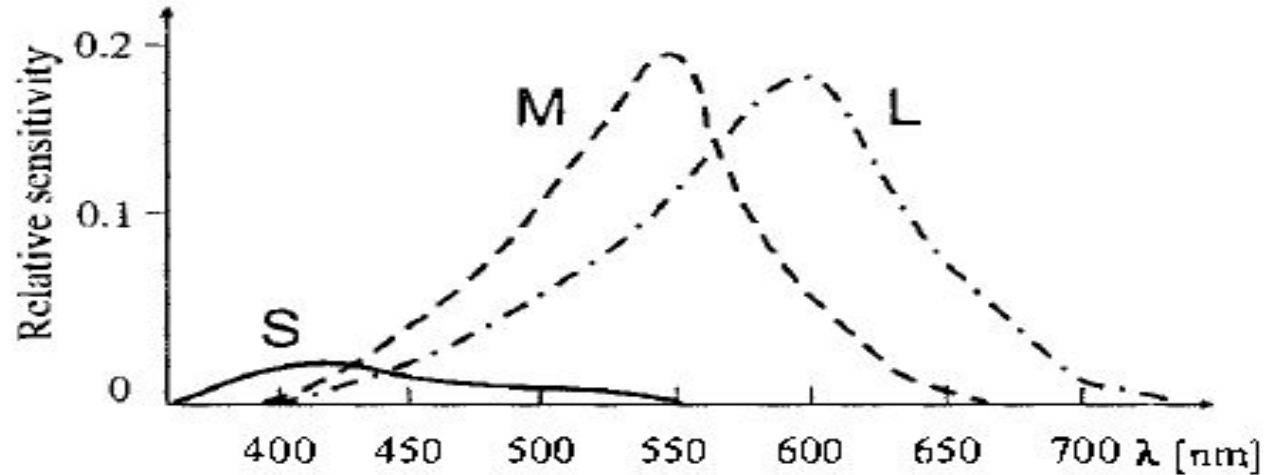
cone-shaped
less sensitive
operate in high light
color vision

Rods (tyčinky sítnice)

rod-shaped
highly sensitive
operate at night
gray-scale vision



Relative sensitivity of S, M, L cones of the human eye



Three types of sensors receptive to the wavelength of incoming irradiation have been established in humans, -> thus the term **trichromacy**.

Trichromacy

Experimental facts:

- Three primaries will work for most people if we allow additive /subtractive matching; “trichromatic” nature of the human visual system
- Most people make the *same* matches for a given set of primaries (i.e., select the same mixtures)
- Color matching experiments imply that three good primaries are sufficient.

Colorimetry

CIE Colorimetric System

Commission Internationale de l'Eclairage

<https://cie.co.at/>



International Commission on Illumination
Commission Internationale de l'Eclairage
Internationale Beleuchtungskommission

The CIE is a technical, scientific and cultural non-profit organization whose objectives are:

INTERNATIONAL
STANDARDS
TECHNICAL
REPORTS....and more

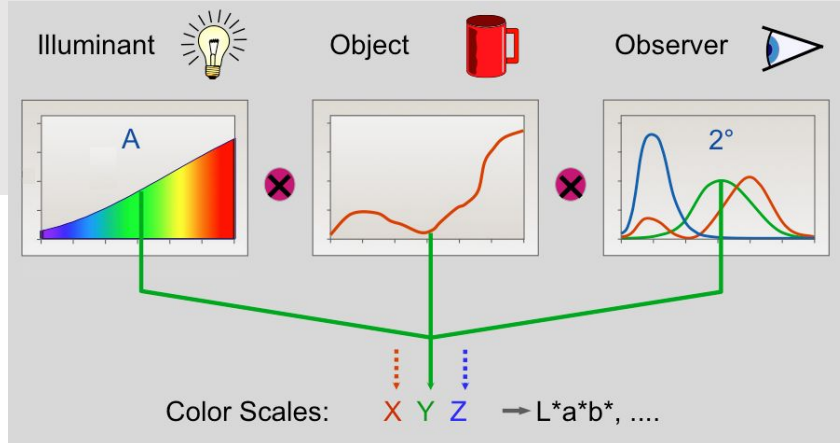
1. To provide an international forum for the discussion of all matters relating to the **science, technology and art in the fields of light and lighting** and for the interchange of information in these fields between countries.
2. To develop **basic standards and procedures of metrology in the fields of light and lighting.**
3. To provide guidance in the application of principles and procedures in the development of international and national standards in the fields of light and lighting.
4. To prepare and publish standards, reports and other publications concerned with all matters relating to science, technology and art in the fields of light and lighting.
5. To maintain liaison and technical interaction with other international organizations concerned with matters related to the science, technology, standardization and art in the fields of light and lighting.

CIE standards

CIE Standard Illuminants

CIE Standard of Reflectance

CIE Standard Colorimetric Observer



CIE Color spaces

CIE standard illuminating and viewing geometries for reflectance measurements.

..and more

CIE standard Illuminants

Common CIE Illuminants

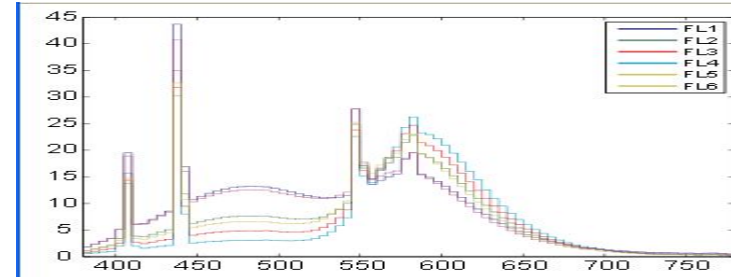
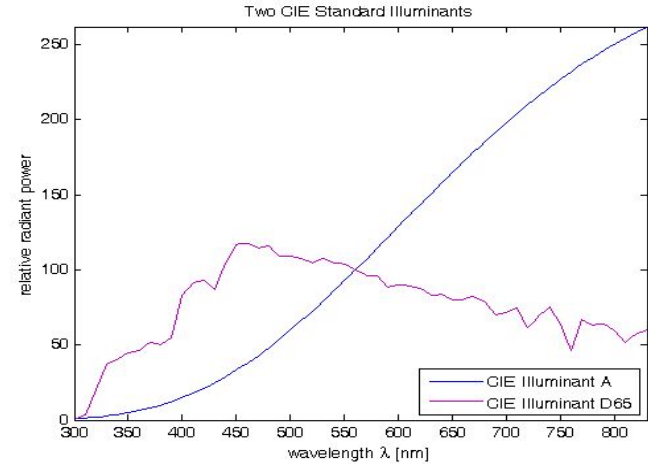
CIE standard illuminant A

tungsten-filament lighting. Its relative spectral power distribution is that of a Planckian radiator at a temperature of approximately 2 856 K.

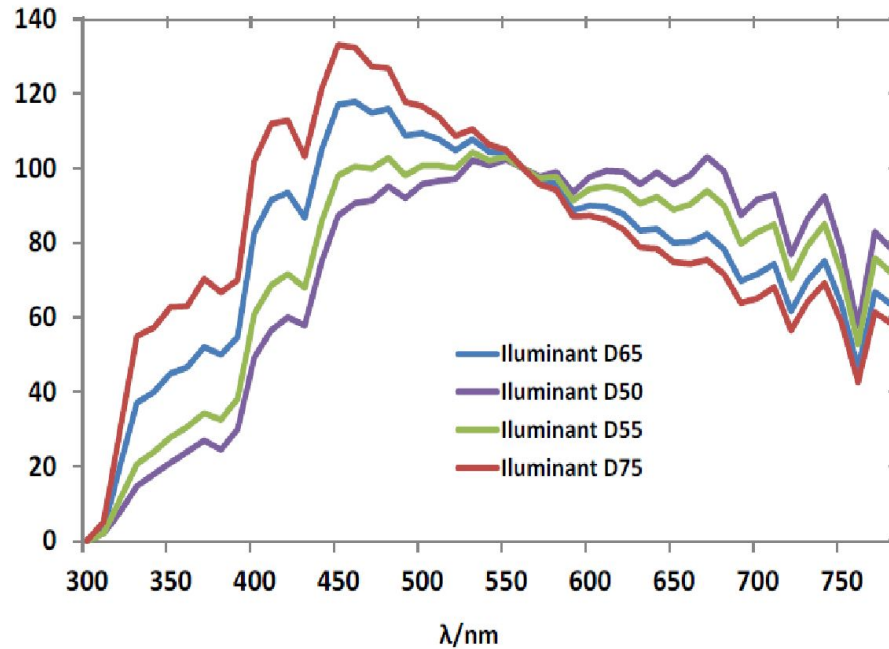
CIE standard illuminant D65

average daylight and has a correlated colour temperature of approximately 6 500 K.

CIE standard illuminant F1-F6 "standard" fluorescent lamps



CIE standard Illuminant - daylight



CIE Standard of Reflectance

CIE Standard of Reflectance

Perfect reflecting diffuser

The CIE (1971) recommends the perfect reflecting diffuser with a reflectance equal to unity.

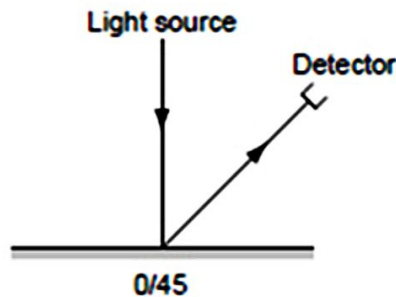
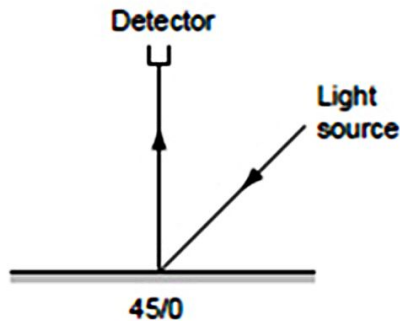
No material surface has those properties

For measurement are used secondary standards - white standards and color standards

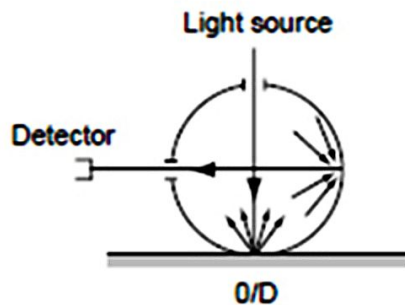
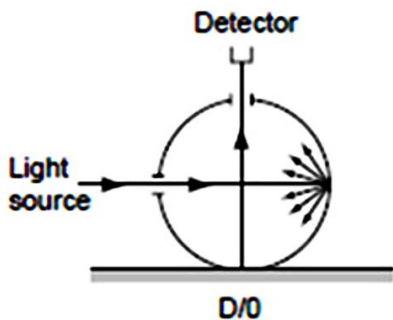


CIE Standard Illuminating and Viewing Conditions

CIE Standard Illuminating and Viewing Conditions

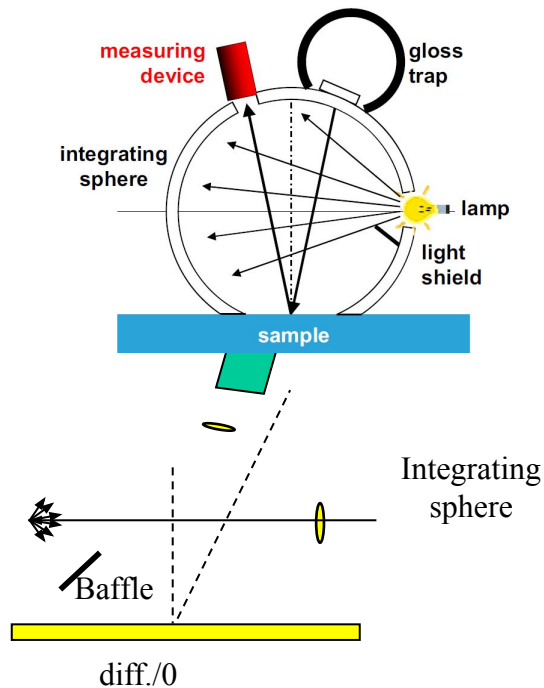


The CIE has recommended 4 types of illumination and viewing geometries for measurement of the object colour:

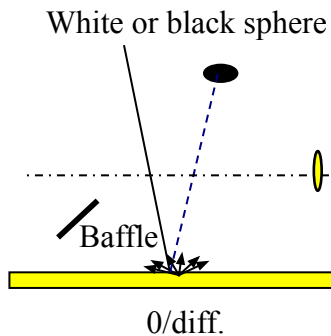


- (a) 45°/normal (45/0)
- (b) normal/45° (0/45)
- (c) diffuse/normal (d/0)
- (d) normal/diffuse (0/d)

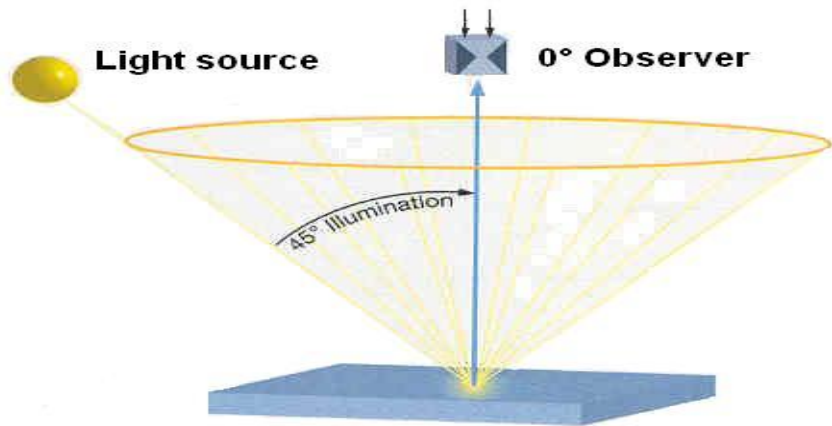
CIE Standard Illuminating and Viewing Conditions



$d/0, 0/d$ Viewing Conditions, $d = 8^\circ$



$45/0, 0/45$ Viewing Conditions



CIE Standard observer color matching functions

CIE 1931 color space

First definition of quantitative links between:

the distributions of wavelengths in the electromagnetic
visible spectrum

and

physiologically perceived colours in human colour
vision.

Commission Internationale de l'Eclairage CIE Colorimetric System

CIE Standard Colorimetric Observer (model of human colour perception):

1931 (2°), 1964 (10°)



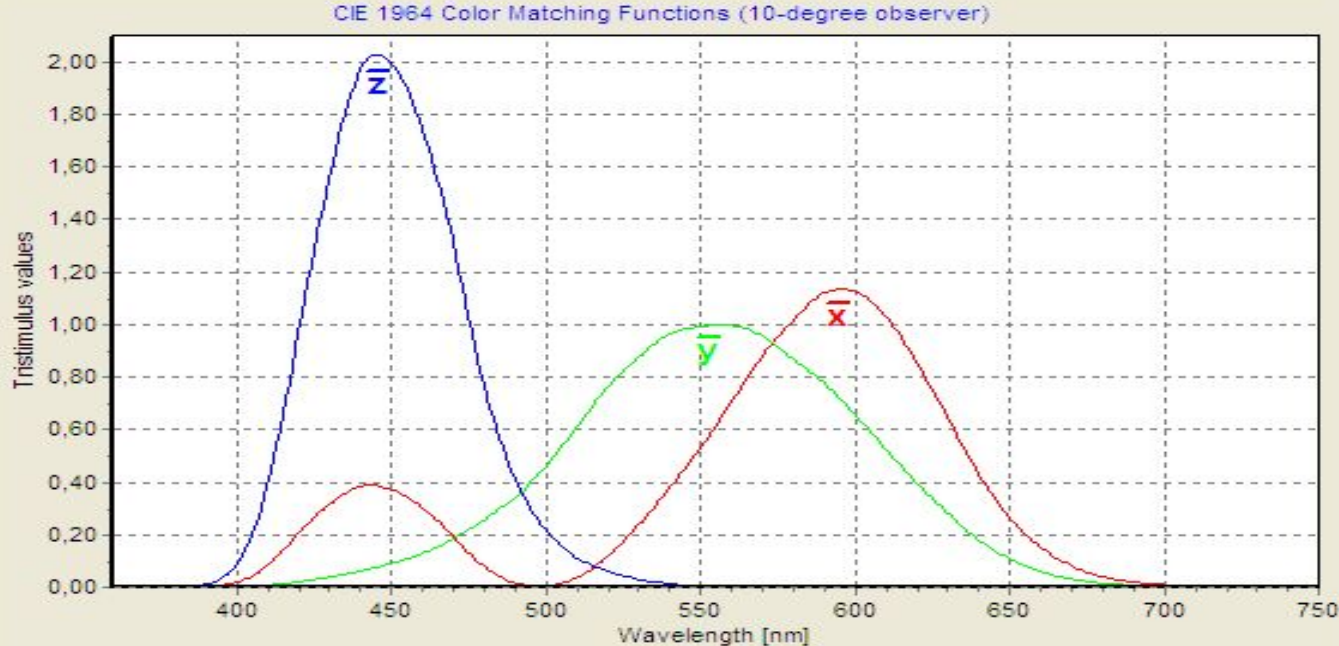
CIE 1931



CIE 1964

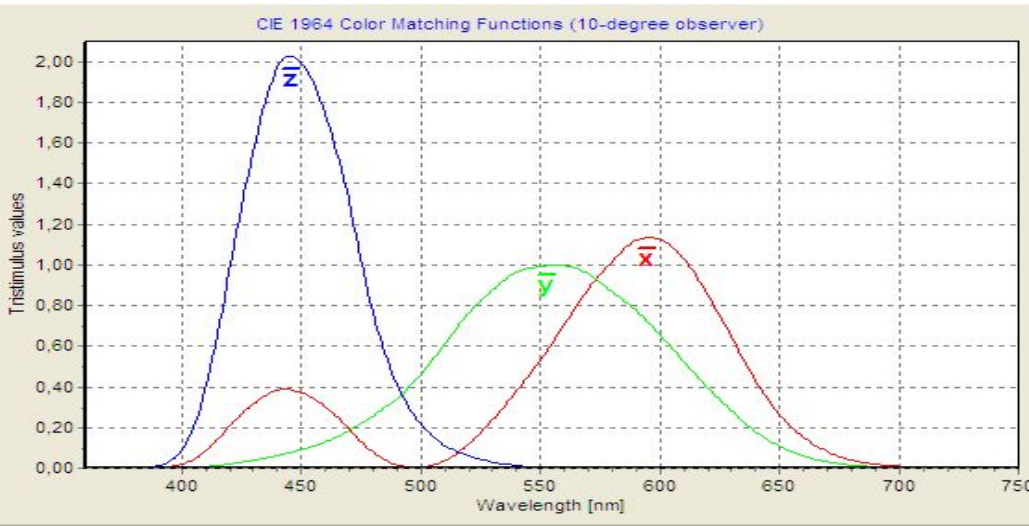
CIE colour matching functions standard colorimetric observer

$x_{10}(\lambda)$, $y_{10}(\lambda)$, $z_{10}(\lambda)$ are colour matching functions that define the CIE 1964
standard colorimetric observer



CIE XYZ color space

X_{10} , Y_{10} , Z_{10} denotes the **tristimulus values** in the CIE 1964:



$$X_{10} = k_{10} \int_{\lambda} P_{\lambda} x_{10}(\lambda) d\lambda$$

$$Y_{10} = k_{10} \int_{\lambda} P_{\lambda} y_{10}(\lambda) d\lambda$$

$$Z_{10} = k_{10} \int_{\lambda} P_{\lambda} z_{10}(\lambda) d\lambda$$

$$k_{10} = \frac{100}{\int_{\lambda} P_{\lambda} y_{10}(\lambda) d\lambda}$$

P_{λ} denotes the **monochromatic component of given color stimulus** with radiant power distribution $P_{\lambda} d\lambda$

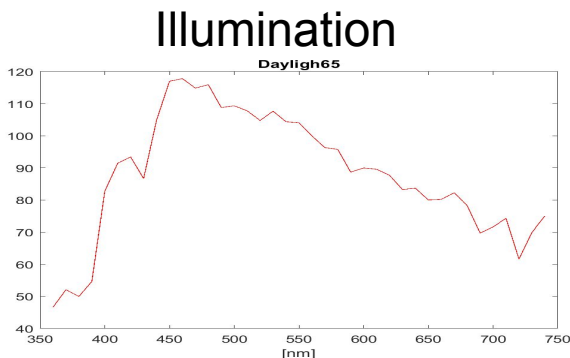
$x_{10}(\lambda)$, $y_{10}(\lambda)$, $z_{10}(\lambda)$ are **color matching functions** that define the CIE 1964 standard colorimetric observer

Color perception model

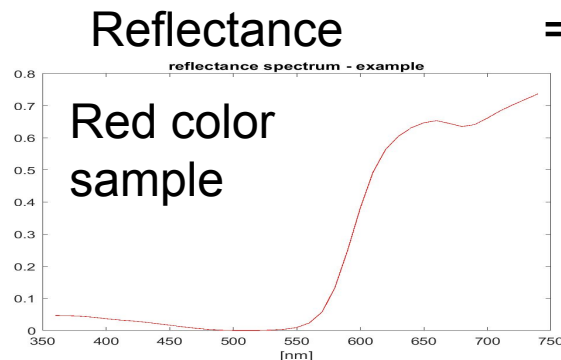
example of CIE XYZ



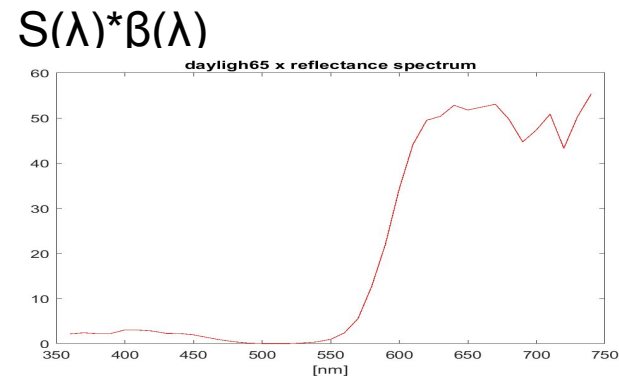
color stimulus with radiant power distribution $P_\lambda d\lambda =$
Illumination $S(\lambda)$ * Reflectance $\beta(\lambda)$



*



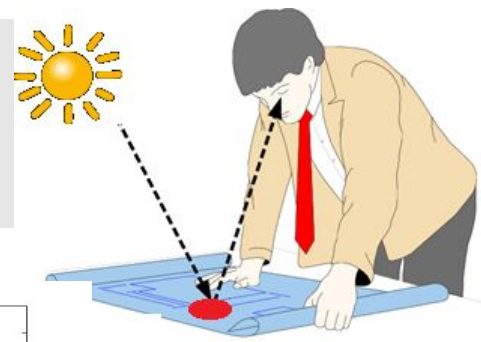
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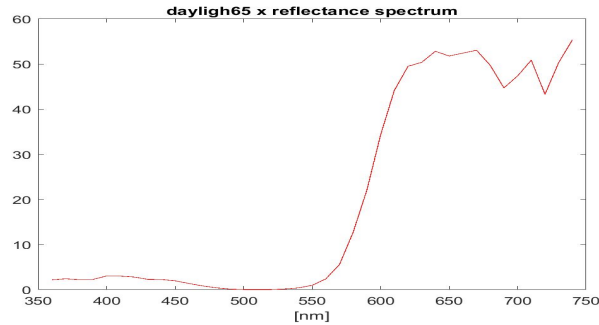
$S(\lambda)$ - spectral power
distribution of the
illuminant

$\beta(\lambda)$ -spectral **reflectance**
factor of the object

Color perception model example of CIE XYZ

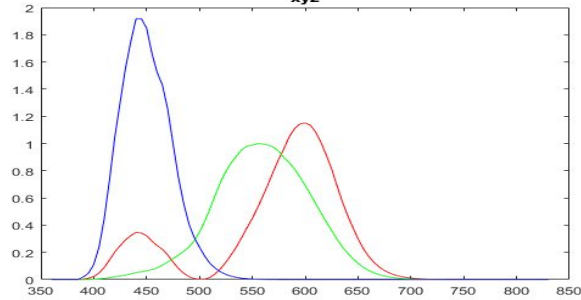


$S(\lambda) * \beta$



*

$\overline{x}_{10}(\lambda) \quad \overline{y}_{10}(\lambda) \quad \overline{z}_{10}(\lambda)$



colour matching functions – **human** - standard
colorimetric observer

$$X = 272.9$$

$$Y = 147.1$$

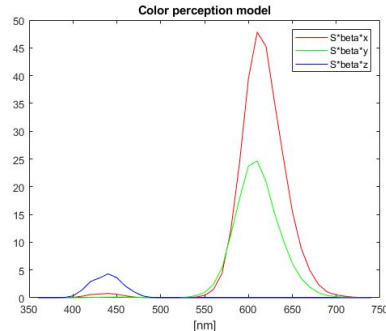
$$Z = 19.6$$

$$X = \sum_{\lambda=380}^{780} S(\lambda) * \beta(\lambda) * \overline{x}_{10}(\lambda)$$

$$Y = \sum_{\lambda=380}^{780} S(\lambda) * \beta(\lambda) * \overline{y}_{10}(\lambda)$$

$$Z = \sum_{\lambda=380}^{780} S(\lambda) * \beta(\lambda) * \overline{z}_{10}(\lambda)$$

=



Color perception model – CIE XYZ color space

$$X = \sum_{\lambda=380\text{nm}}^{780\text{nm}} S(\lambda) * \beta(\lambda) * x(\lambda)$$

$$Y = \sum_{\lambda=380\text{nm}}^{780\text{nm}} S(\lambda) * \beta(\lambda) * y(\lambda)$$

$$Z = \sum_{\lambda=380\text{nm}}^{780\text{nm}} S(\lambda) * \beta(\lambda) * z(\lambda)$$

$\overline{x}_{10}(\lambda)$ $\overline{y}_{10}(\lambda)$ $\overline{z}_{10}(\lambda)$ colour matching functions → **human** - standard colorimetric observer

$S(\lambda)$ spectral power distribution of the **illuminant**

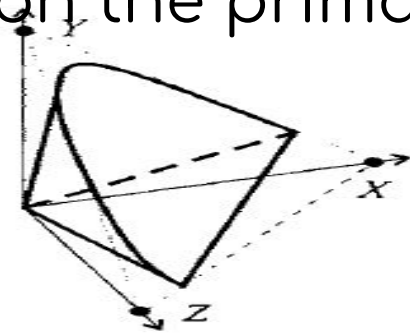
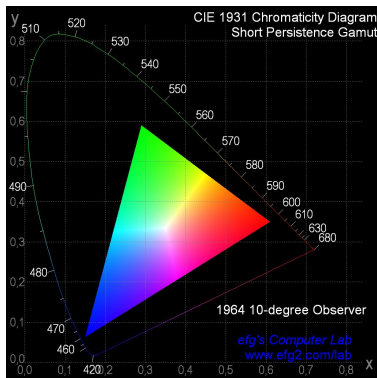
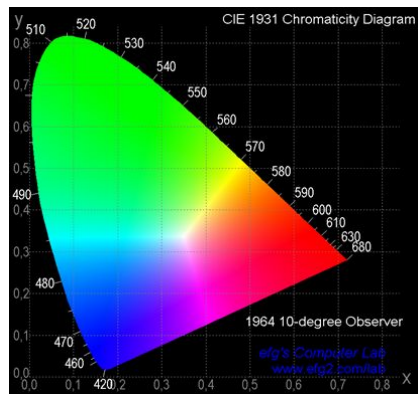
$\beta(\lambda)$ spectral **reflectance** factor of the object

CIE Chromaticity diagram

CIE Chromaticity coordinates

Normalized tristimulus values, called **chromaticity coordinates**, are calculated based on the primaries as follows:

CIE chromaticity diagram:

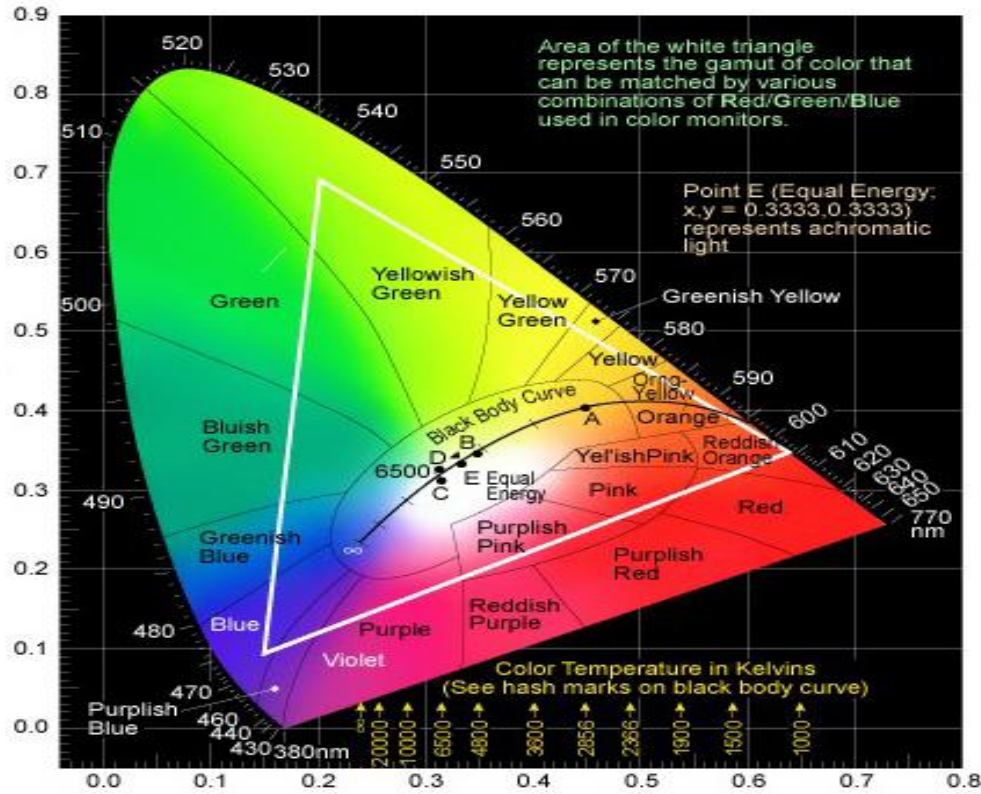


$$x_{10} = \frac{X_{10}}{X_{10} + Y_{10} + Z_{10}}$$

$$y_{10} = \frac{Y_{10}}{X_{10} + Y_{10} + Z_{10}}$$

CIE Chromaticity coordinates

Chromaticity diagram

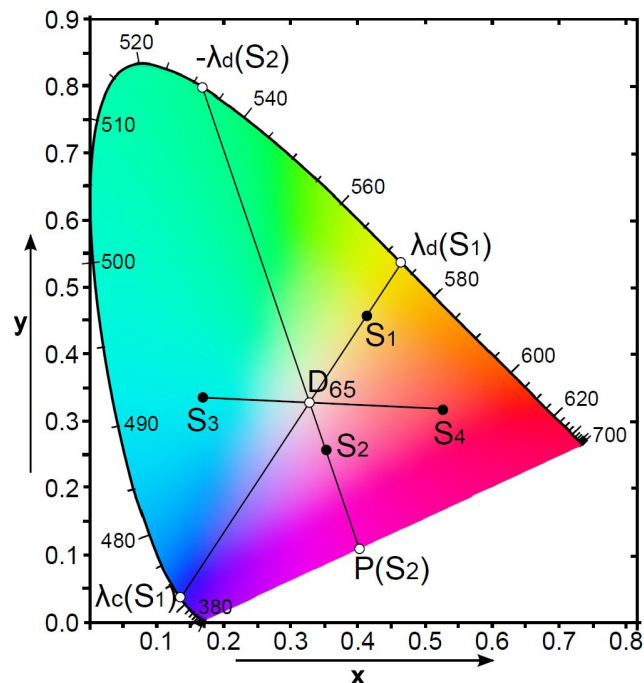


color gamut is a certain *complete subset* of colors
- for instance the white triangle

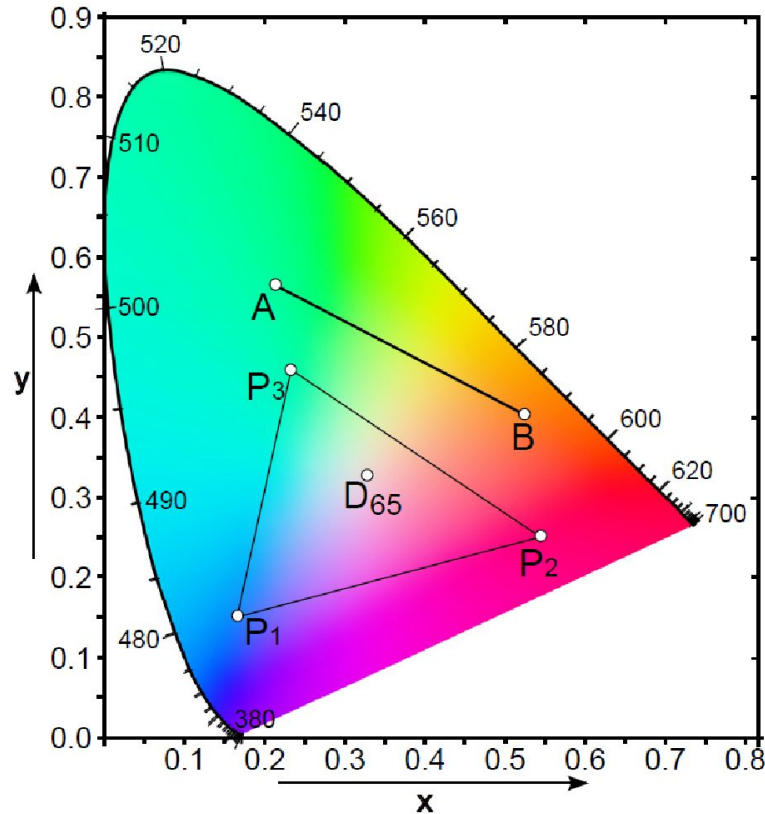
CIE Chromaticity diagram

The wavelength of the monochromatic light at the point of intersection (point D) with the horseshoe-shaped curve is termed the **dominant wavelength**, λ_d , for the respective test colour.

The ratio of the lines D65-S2 and D65P(S2) provides a measure called **excitation purity**, similar to saturation.

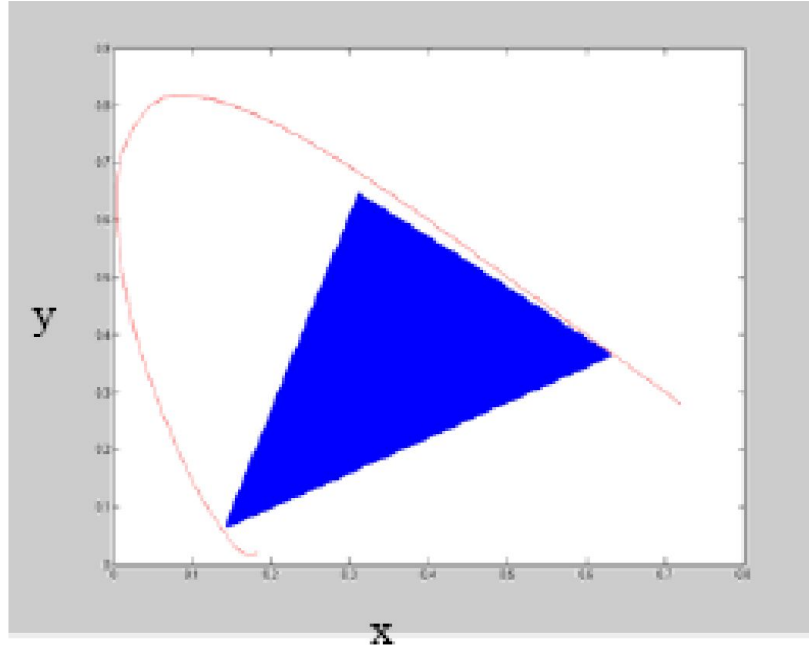


Chromaticity diagram

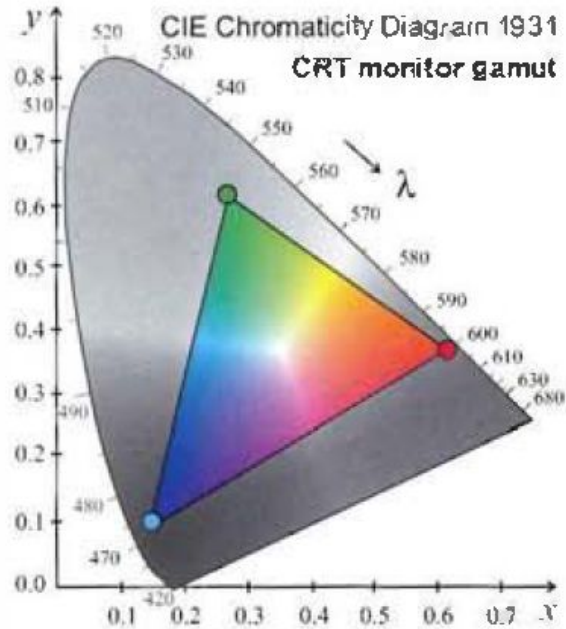


Chromaticity diagram & Gamut of camera

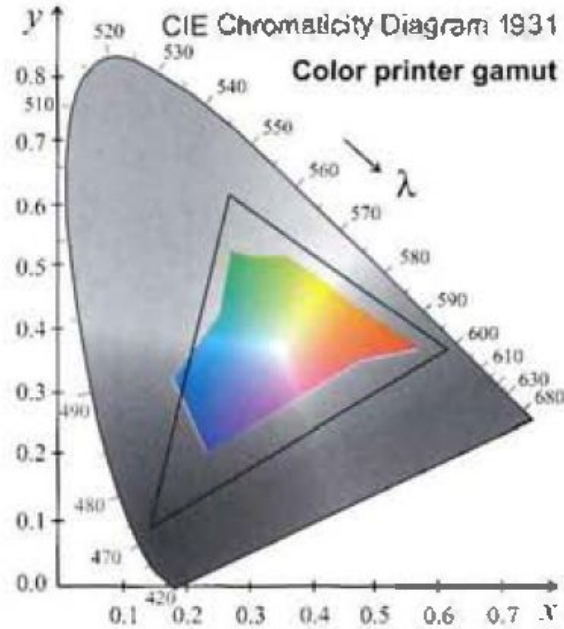
The blue triangle in the Figure restricts the range of colors which can be reproduced by a RGB camera.



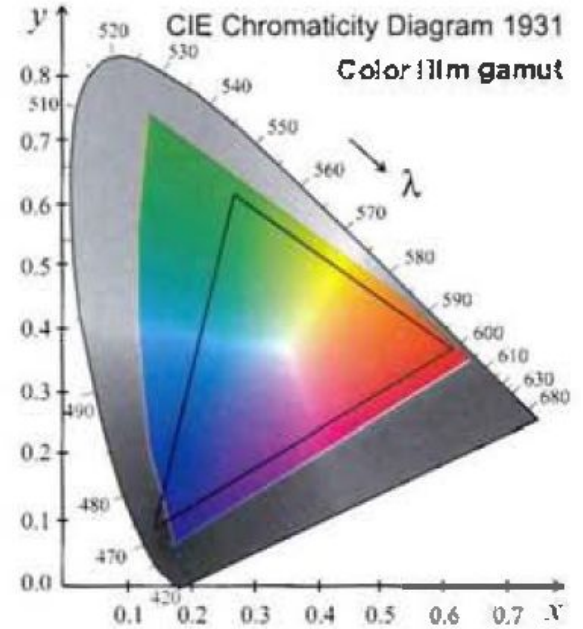
Gamuts of three typical display devices



(a) CRT monitor



(b) printer



(c) film

Color matching

Color matching

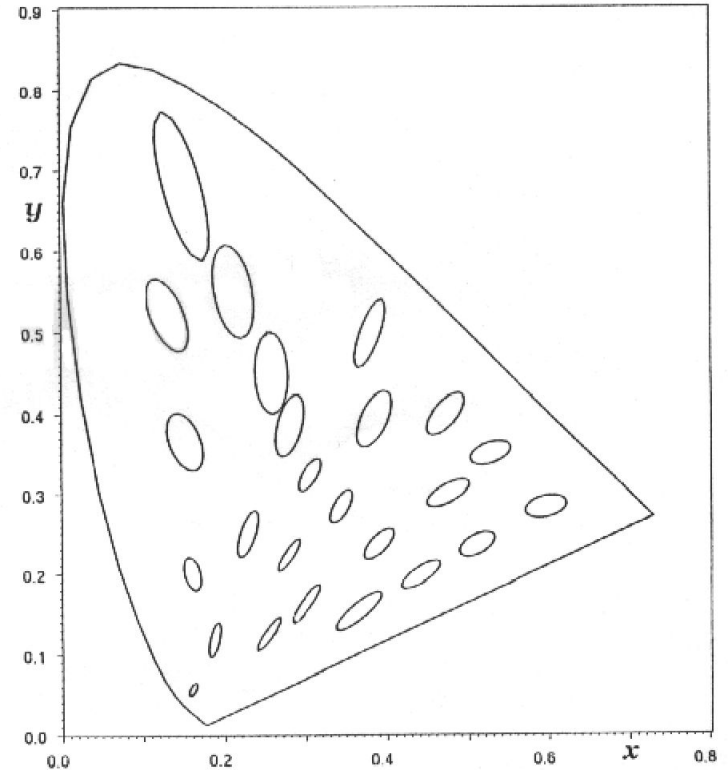
Matching two light patches to complete colour match:

lightness, hue and saturation.

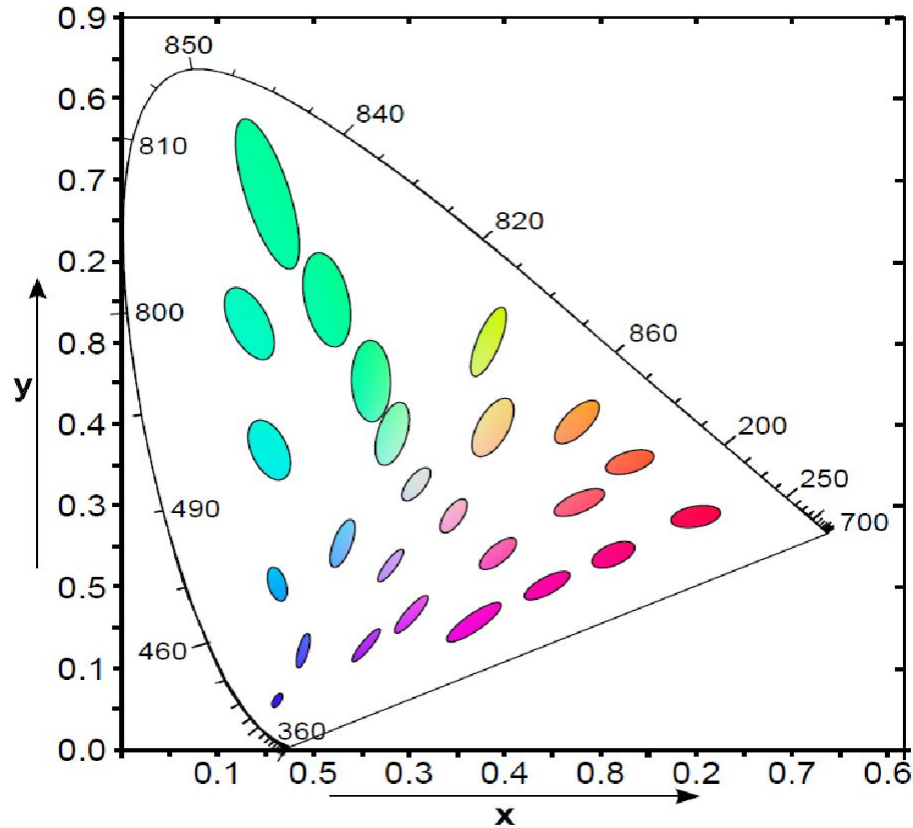
Color differences

For the fovea vision, related to a visual field of 2° , the non uniformity of the chromatic scale has been measured by MacAdam and is graphically represented by ellipses on the chromaticity diagram.

MacAdam ellipses on the CIE 1931 chromaticity diagram magnified 10 times



MacAdam ellipses



Uniform color space

CIE L*a*b*

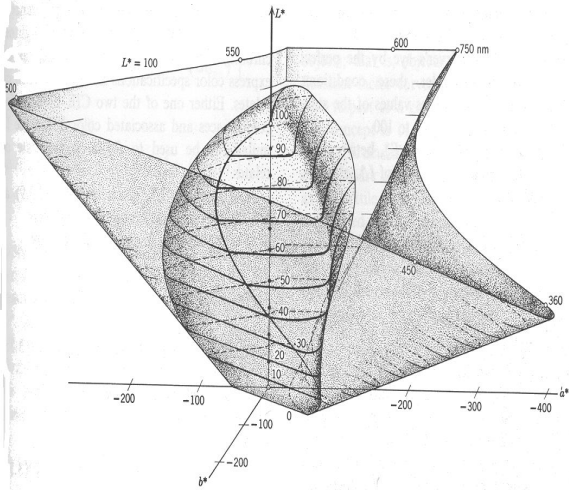
Absolute value of the L*a*b* euclidean colour differences corresponds to the human perception of the colour differences

Definition:

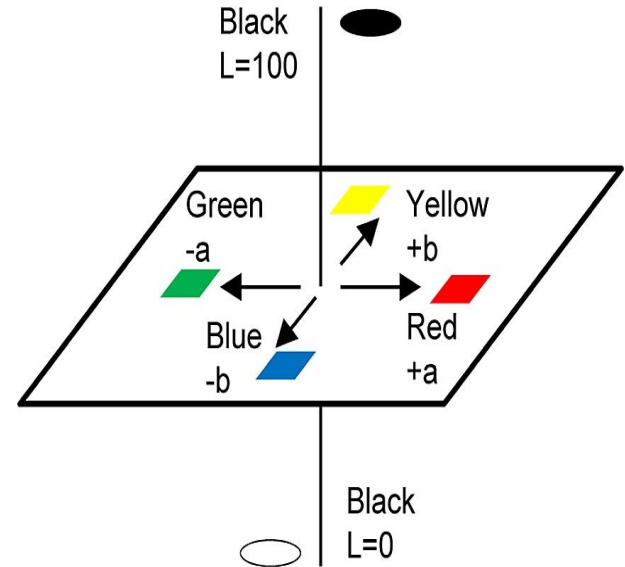
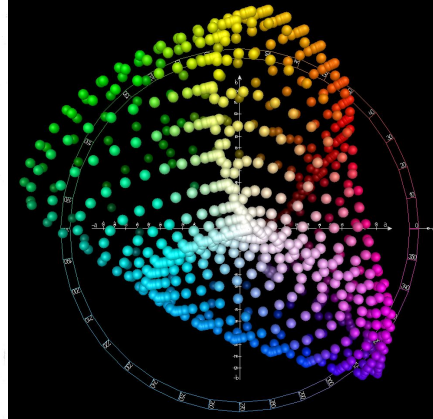
$$\begin{aligned} L^* &= 116 \left(\frac{Y}{Y_n} \right)^{\frac{1}{3}} - 16 & \text{if } \frac{Y}{Y_n} > 0.008856 & \quad a^* = 500 \left[\left(\frac{X}{X_n} \right)^{\frac{1}{3}} - \left(\frac{Y}{Y_n} \right)^{\frac{1}{3}} \right] \\ L^* &= 903.3 \left(\frac{Y}{Y_n} \right)^{\frac{1}{3}} & \text{if } \frac{Y}{Y_n} \leq 0.008856 & \quad b^* = 200 \left[\left(\frac{Y}{Y_n} \right)^{\frac{1}{3}} - \left(\frac{Z}{Z_n} \right)^{\frac{1}{3}} \right] \end{aligned}$$

where $[X_n, Y_n, Z_n]$ is the white reference point.

Uniform color space CIE $L^*a^*b^*$



Sketch of CIE
1976 $L^*a^*b^*$
color space



Colour Differences – Euclidean distance in $L^*a^*b^*$ Colour space

The total colour difference ΔE^* between two color stimuli: L^*_1, a^*_1, b^*_1 and L^*_2, a^*_2, b^*_2 :

$$\Delta L^* = L^*_1 - L^*_2$$

$$\Delta a^* = a^*_1 - a^*_2$$

$$\Delta b^* = b^*_1 - b^*_2$$

CIE 1976 ($L^*a^*b^*$) colour difference formula

$$\Delta E^*_{ab} = \sqrt{[(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2]}$$

Metamerism

Object-colour stimuli of different spectral radiant power distributions and the given observer judges the two stimuli to be in complete colour match.

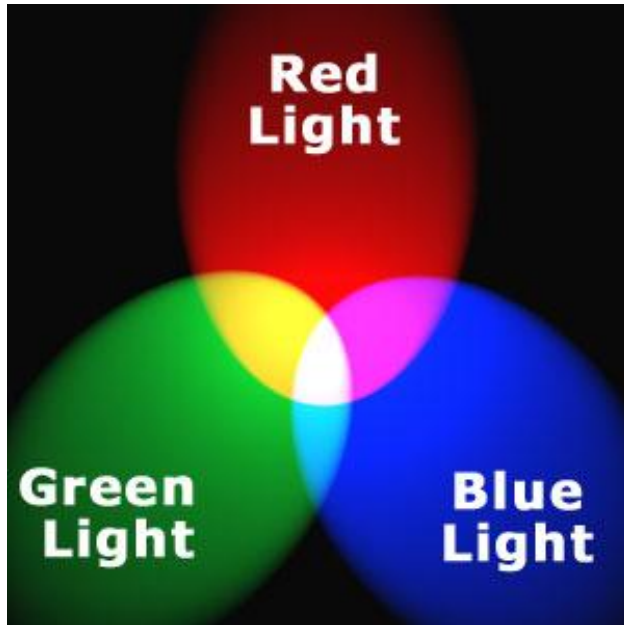
The colour match cannot be expected to remain a colour match if the illuminant is changed to another one of different spectral radiant power distribution

Other color spaces

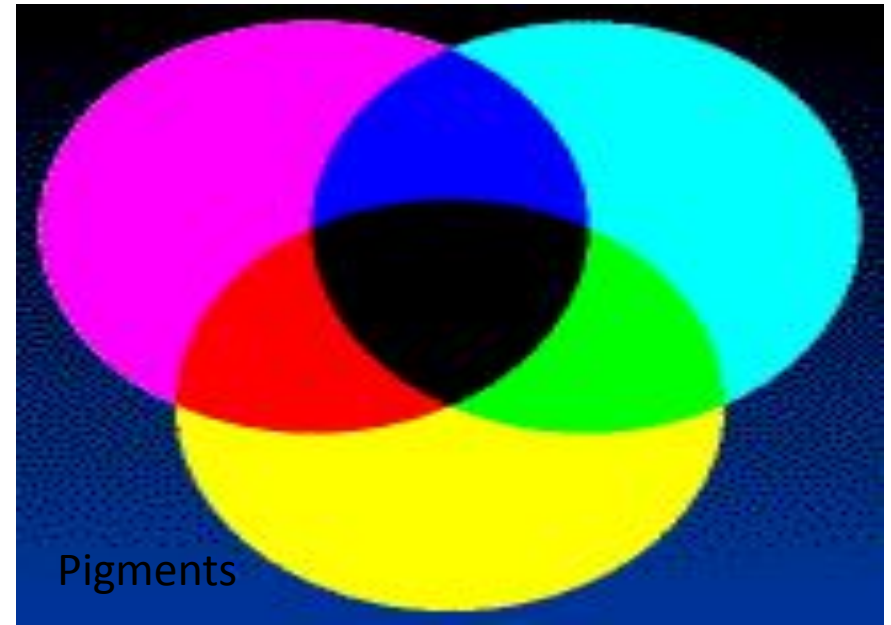
Additive color mixing

Subtractive color mixing

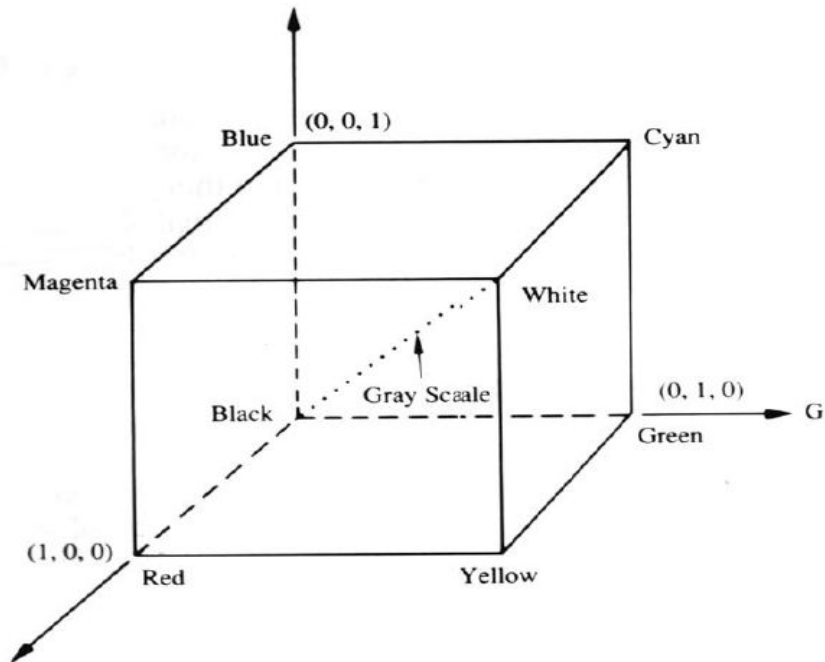
Additive color mixing



Subtractive color mixing



sRGB color spaces



RGB to XYZ – example a camera (TVI)

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 3.2405 & -1.5372 & -0.4985 \\ -0.9693 & 1.8760 & 0.0416 \\ 0.0556 & -0.2040 & 1.0573 \end{bmatrix} * \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

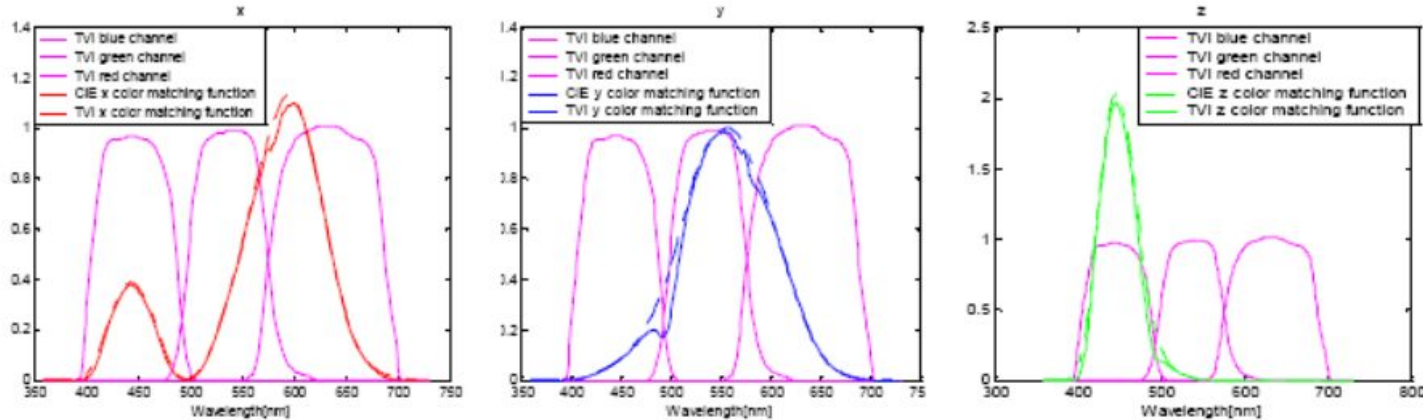


Figure 2.1: CIE color matching functions compared with the color matching functions calculated for the camera

RGB to XYZ

variety of definitions

Adobe RGB (1998)	D65	0.5767309 0.2973769 0.0270343	0.1855540 0.6273491 0.0706872	0.1881852 0.0752741 0.9911085	2.0413690 -0.9692660 0.0134474	-0.5649464 1.8760108 -0.1183897	-0.3446944 0.0415560 1.0154096
AppleRGB	D65	0.4497288 0.2446525 0.0251848	0.3162486 0.6720283 0.1411824	0.1844926 0.0833192 0.9224628	2.9515373 -1.0851093 0.0854934	-1.2894116 1.9908566 -0.2694964	-0.4738445 0.0372026 1.0912975
Best RGB	D50	0.6326696 0.2284569 0.0000000	0.2045558 0.7373523 0.0095142	0.1269946 0.0341908 0.8156958	1.7552599 -0.5441336 0.0063467	-0.4836786 1.5068789 -0.0175761	-0.2530000 0.0215528 1.2256959
Beta RGB	D50	0.6712537 0.3032726 0.0000000	0.1745834 0.6637861 0.0407010	0.1183829 0.0329413 0.7845090	1.6832270 -0.7710229 0.0400013	-0.4282363 1.7065571 -0.0885376	-0.2360185 0.0446900 1.2723640
Bruce RGB	D65	0.4674162 0.2410115 0.0219101	0.2944512 0.6835475 0.0736128	0.1886026 0.0754410 0.9933071	2.7454669 -0.9692660 0.0112723	-1.1358136 1.8760108 -0.1139754	-0.4350269 0.0415560 1.0132541
CIE RGB	E	0.4887180 0.1762044 0.0000000	0.3106803 0.8129847 0.0102048	0.2006017 0.0108109 0.9897952	2.3706743 -0.5138850 0.0052982	-0.9000405 1.4253036 -0.0146949	-0.4706338 0.0885814 1.0093968

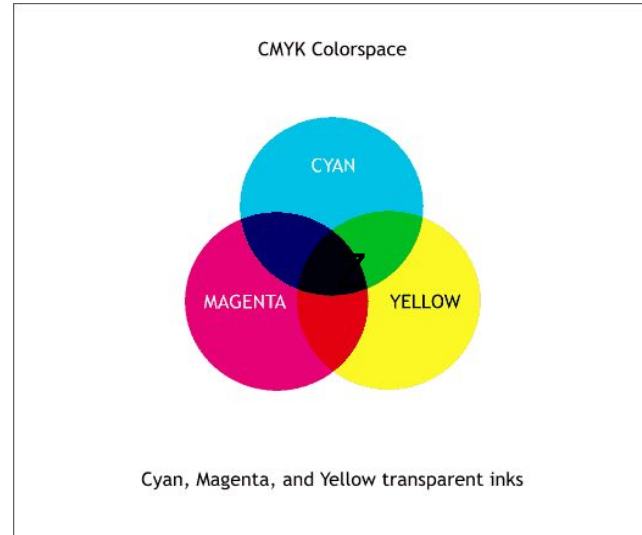
..and more...

CMY colour model

Cyan, Magenta, Yellow

- subtractive colour mixing which is used in printing processes

CMYK stores ink values for black in addition.



Color Space – YCbCr

Conversion from RGB:

$$Y = 0.299(R - G) + G + 0.114(B - G)$$

$$Cb = 0.564(B - Y)$$

$$Cr = 0.713(R - Y)$$

Advantage: Bandwidth efficiency

Used for: digital video encoding

Axes:

Y: luma

Cb: blue chroma

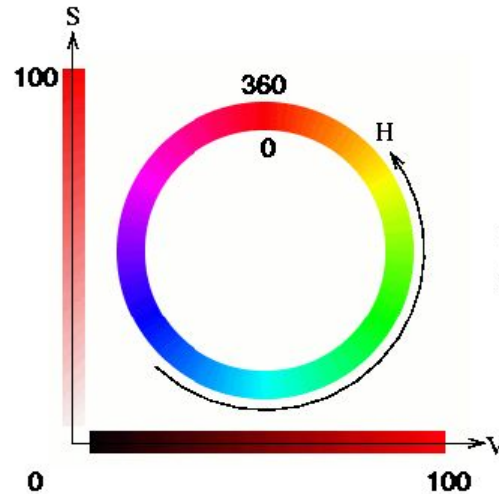
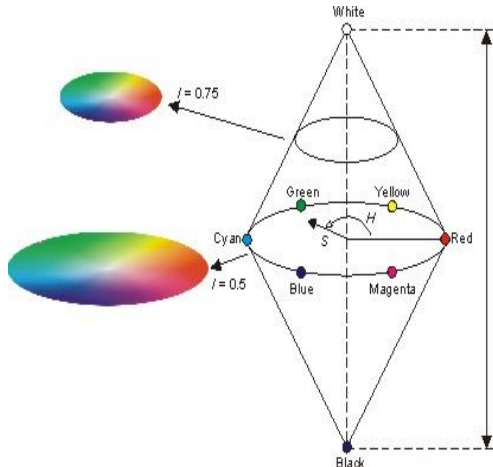
Cr: red chroma

HSI color space

saturation (S) is proportional to radial distance

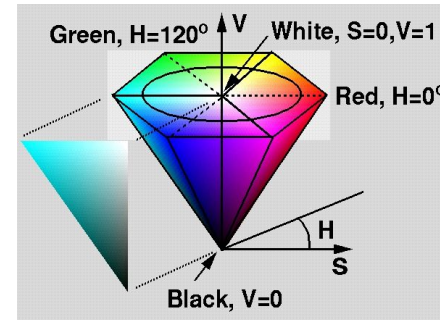
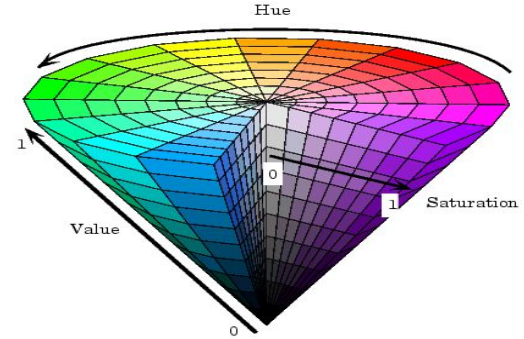
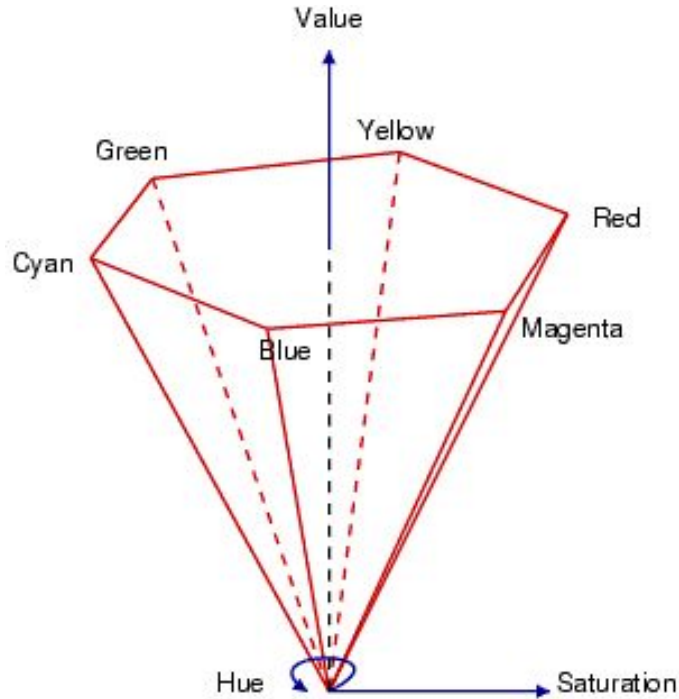
hue (H) is a function of the angle in the polar coordinate system.

Intensity(I): $I = (R+G+B)/3$



HSV colour space

Hue, Saturation, and Value



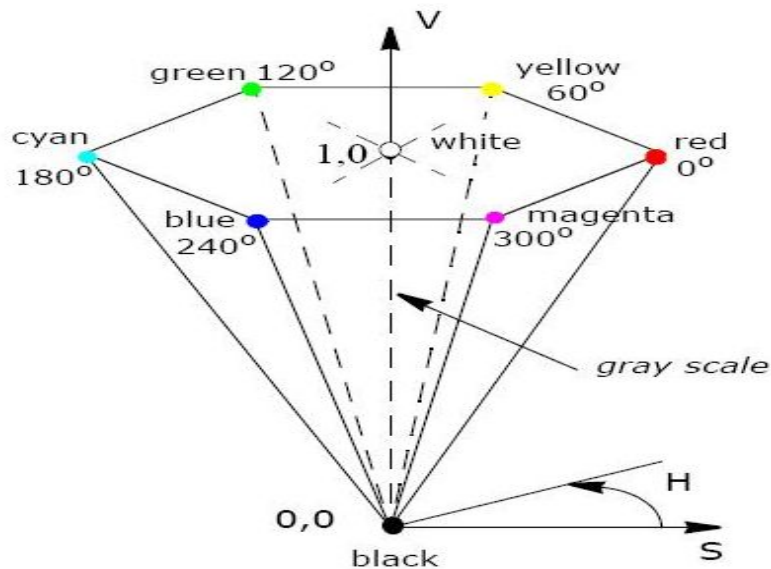
- often used by painters

HLS

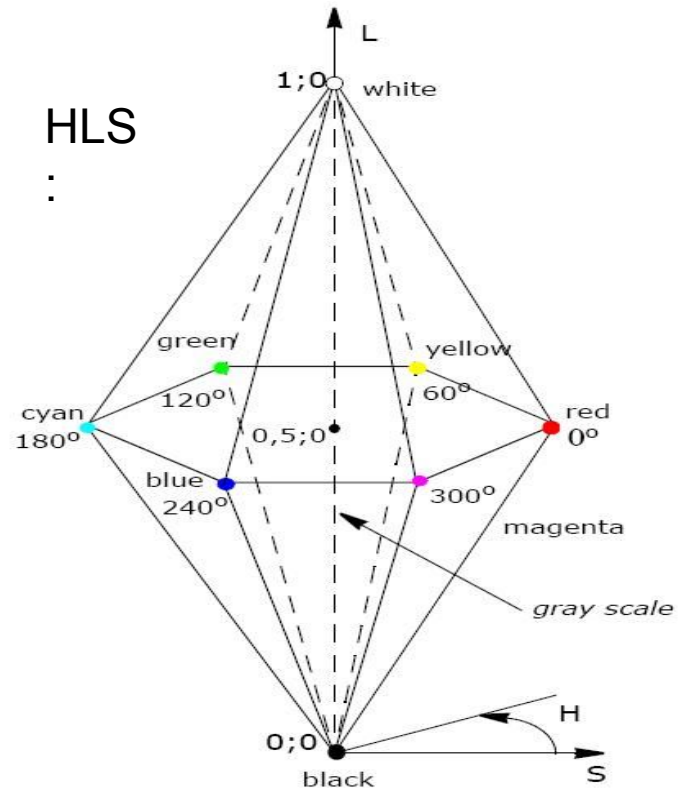
hue, lightness, saturation

HSV vs. HLS

HSV
:



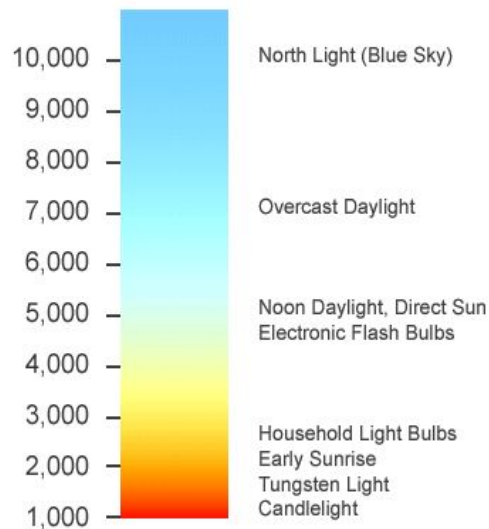
HLS
:



Correlated Colour Temperature (CCT) of light sources

is defined as the value of the temperature of the black body radiator when the radiator colour matches that of the light source

Colour Temperatures in the Kelvin Scale



1500 K	Candlelight
2680 K	40-watt lamp
3200 K	Sunrise/sunset
3400 K	One hour from dusk/dawn
5000-4500 K	Xenon lamp/light arc
5500 K	Sunny daylight around noon
5500-5600 K	Electronic photo flash
6500-7500 K	Overcast sky
9000-12000 K	Blue sky

Measurement of color

Minolta 2600d Photospectrometer

This instrument utilizes the $d/8^\circ$ geometry conforming to CIE standard.

Illumination / Viewing system $d/8^\circ$
(diffuse illumination, 8-degree viewing)

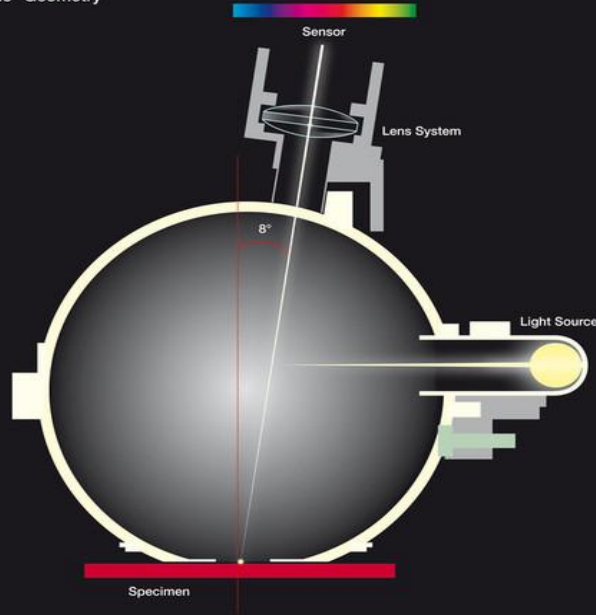
specular component included (SCI) and excluded (SCE).



Diffuse illumination - d/8° geometry

Ulbricht sphere

di:8° Geometry

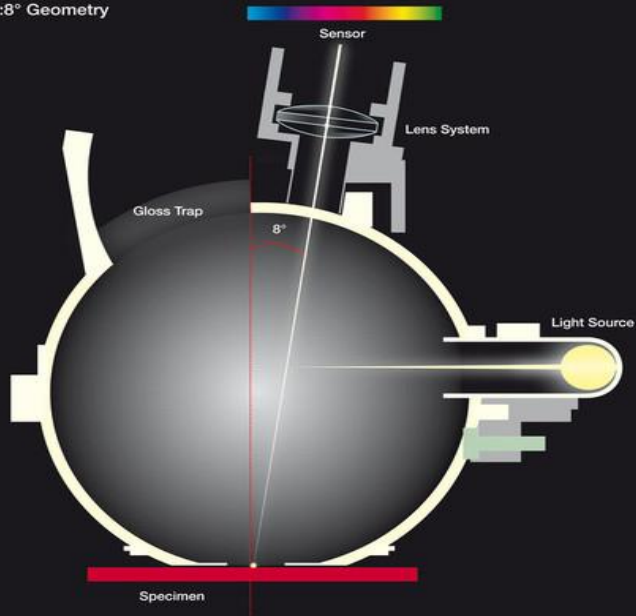


- The light e.g. of a xenon lamp is projected into a sphere.
- The interior of the sphere is coated with a white highly reflecting substance (barium sulphate, ceramic, special plastic) which reflects the light manifold.
- The sample is positioned at an opening of the sphere and is illuminated from all directions with a close to perfect diffuse light.

Diffuse illumination - d/8° geometry

Ulbricht sphere

de:8° Geometry



- In order to prevent reflection of specular light from the sample surface, many instruments feature a **gloss trap**.

Examples

Wood quality measurement

Examples



0 Tage



30 Tage



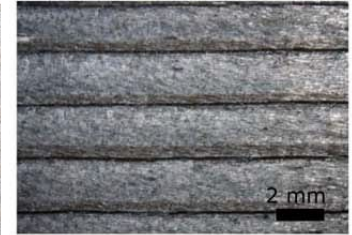
95 Tage



166 Tage



270 Tage



366 Tage